

Hybridisation (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Analyse the hybridisation and bond formation in molecules and ions.
- Identify the relationships between Lewis formulas, electron domains, molecular geometry and type of hybridisation.
- Predict the geometry around an atom from its sp , sp^2 and sp^3 hybridisation, and vice versa for both organic and inorganic examples.

Consider the element carbon, which has been discussed a lot over the course of this subtopic.

Remember from [section S1.3.5 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) that its atomic number of 6 will give an electron configuration of $1s^2, 2s^2, 2p_x^1, 2p_y^1$. The 2s-orbital can hold a maximum of two electrons, which carbon has (**Figure 1**). This would indicate that carbon has **two** singly occupied orbitals ($2p_x$ and $2p_y$) that are available for bonding, and so you may expect carbon to form only two bonds.

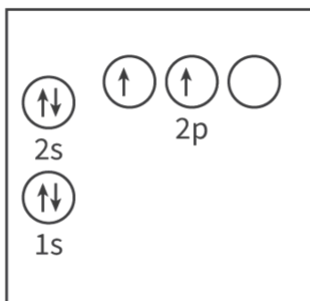


Figure 1. Electron configuration of carbon.

 More information for figure 1

The image shows a diagram of the electron configuration for a carbon atom. It displays two rows of orbital notations. The upper row represents the 2p orbitals, showing three circles labeled '2p' with two singly occupied by upward arrows, indicating unpaired electrons in the 2p_x and 2p_y orbitals. The lower row shows a filled circle labeled '2s' with one pair of arrows (upward and downward) indicating a filled orbital, and below that, a filled circle labeled '1s.' These depictions reflect carbon's electron configuration, highlighting its two unpaired electrons available for bonding in its 2p_x and 2p_y orbitals.

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However, each of the carbon-based molecules and networks covered throughout this subtopic do not support this: carbon has typically always formed four bonds.

In methane, CH₄, carbon forms four single bonds to hydrogens.

To form diamond, carbon forms four covalent bonds to other carbons, which join to other carbons in a covalent network structure. In both these covalent species, each central carbon has four electron domains with a tetrahedral molecular geometry and a bond angle of 109.5°.

In a molecule of carbon dioxide, CO₂, carbon forms two double bonds to oxygen atoms. The two electron domains give it a linear molecular geometry with a bond angle of 180° (**Figure 2**).

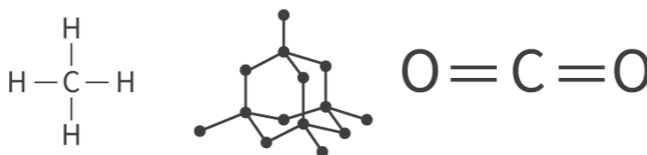


Figure 2. Carbon-based molecules and networks.

 More information for figure 2

The image shows a representation of carbon-based molecules and networks. It includes three different molecular structures:

1. On the left, there is a structural formula of methane (CH₄), where a carbon atom is centrally bonded to four hydrogen atoms in a tetrahedral arrangement.
2. In the center, a hexagonal ring structure likely represents a generic carbon network such as benzene or similar cyclic compound, with each vertex representing carbon atoms. Some bonds extend vertically, indicating possible connections with other molecules or atoms.
3. On the right, the linear molecular structure of carbon dioxide (CO₂) is depicted, with double bonds between the central carbon atom and the two oxygen atoms, giving it a linear geometry.

The image demonstrates the versatility of carbon in forming different types of bonds and structures, from simple hydrocarbons to complex networks.

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However, all these structures indicate that carbon actually forms four bonds. It must have **four** orbitals that get involved in its bonding. How is this possible?

Excitation

Carbon's lowest energy or ground state electron configuration changes during bonding during a process called excitation (**Figure 3**). This is the process of changing from a ground state to an excited state by energy absorption.

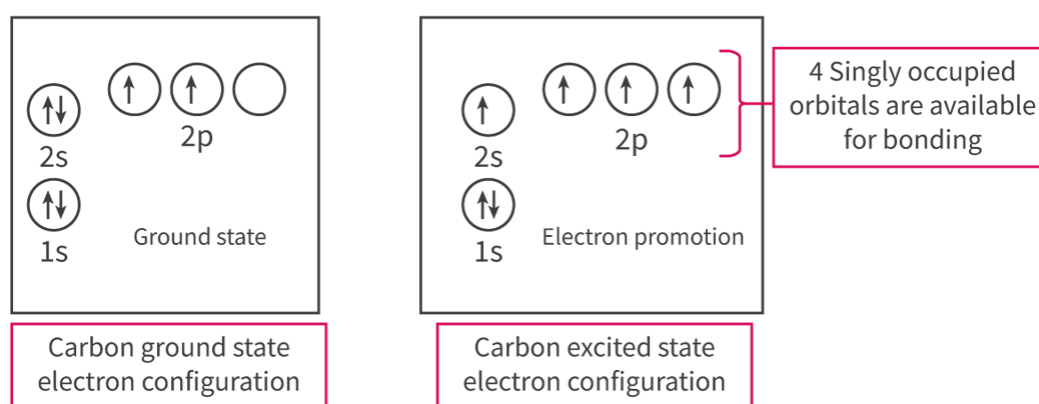


Figure 3. Excitation of carbon's electrons.

 More information for figure 3

The diagram illustrates the excitation of carbon's electrons from the ground state to an excited state, which is crucial for bonding. On the left, the ground state electron configuration is shown. It includes two 1s electrons represented by downward and upward arrows in a circle, and two 2s electrons shown with an upward arrow and a downward arrow in another circle. Additionally, the 2p orbital has two upward arrows in separate circles symbolizing unoccupied and half-filled orbitals, labeled as "Ground state."

On the right side of the diagram, the excited state of electron configuration in carbon is depicted, labeled as "Electron promotion." Here, there are two electrons in the 1s orbital (one upward arrow, one downward arrow), and the 2s orbital contains a single upward arrow, indicating excitation. The 2p orbital now has three upward arrows spread across three circles, showing that the additional electron has moved. The area is annotated with "4 Singly occupied orbitals are available for bonding," highlighting the preparation for forming covalent bonds. The bottom of the diagram specifies "Carbon ground state electron configuration" and "Carbon excited state electron configuration."

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Making connections

In sections S1.3.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>) and S1.3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>) the movement of hydrogen's electrons from higher energy levels to lower energy levels is discussed, along with how this produces an emission spectrum as energy is emitted. The concept of excitation works off a similar, but opposite principle.

Consider the structures of methane, diamond and carbon dioxide we have just analysed; they all seem to indicate that the four bonds formed would be equal. However, the atomic orbitals in carbon's excited state that are available for bonding are not all the same; there is one s-orbital and three p-orbitals at slightly higher energy. This suggests that these orbitals have somehow must have been changed and made equal during the bonding process.

sp^3 hybridisation

A helpful theory that can explain this 'equalising' of bonding orbits describes a process of hybridisation. This is the process by which atomic orbitals within an atom mix to produce hybrid orbitals as seen in **Figure 4**.

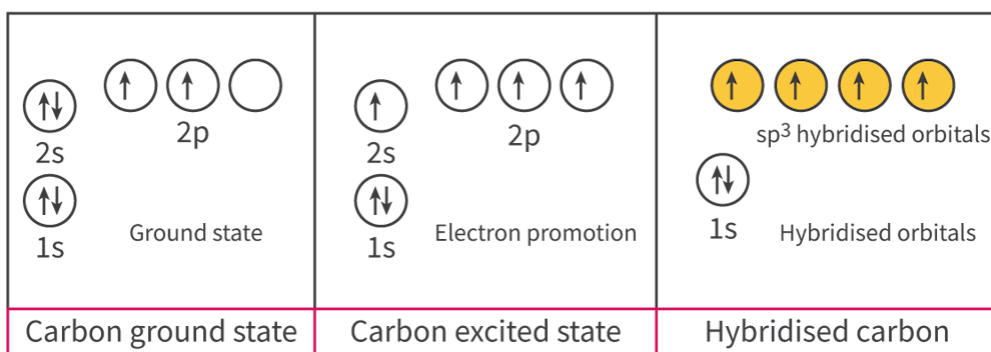


Figure 4. sp^3 hybridisation of carbon's orbitals.

 More information for figure 4

The diagram illustrates the sp^3 hybridisation of carbon's orbitals in three stages: the ground state, excited state, and hybridised state. The first section represents the carbon ground state, showing two electrons in the 1s orbital and two electrons each in one of the 2s and 2p orbitals. The second section, labeled 'Carbon excited state,' indicates electron promotion, showing that one of the electrons from the 2s orbital is shifted to an empty 2p orbital. Finally, the third section shows 'Hybridised carbon,' where four sp^3 hybridised orbitals

are formed, each containing one electron. The diagram effectively highlights the redistribution and mixing of electrons to form four equivalent hybrid orbitals.

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These hybrid orbitals have different shapes, orientations in space and energies than their parent orbitals, and can form stronger bonds with greater overlap. Note that the total energy of the orbitals has not changed, it has just been redistributed equally among four new hybrid orbitals that are now at intermediate energy level. The hybridisation has made the molecule more stable.

The new shape and orientations in space of these hybrid orbitals can be seen in **Figure 5**. You can see the new bond angle that results from this hybridisation; spherical s-orbitals and dumb-bell-shaped p-orbitals which were at 90° to each other can now form bond angles of 109.5° .

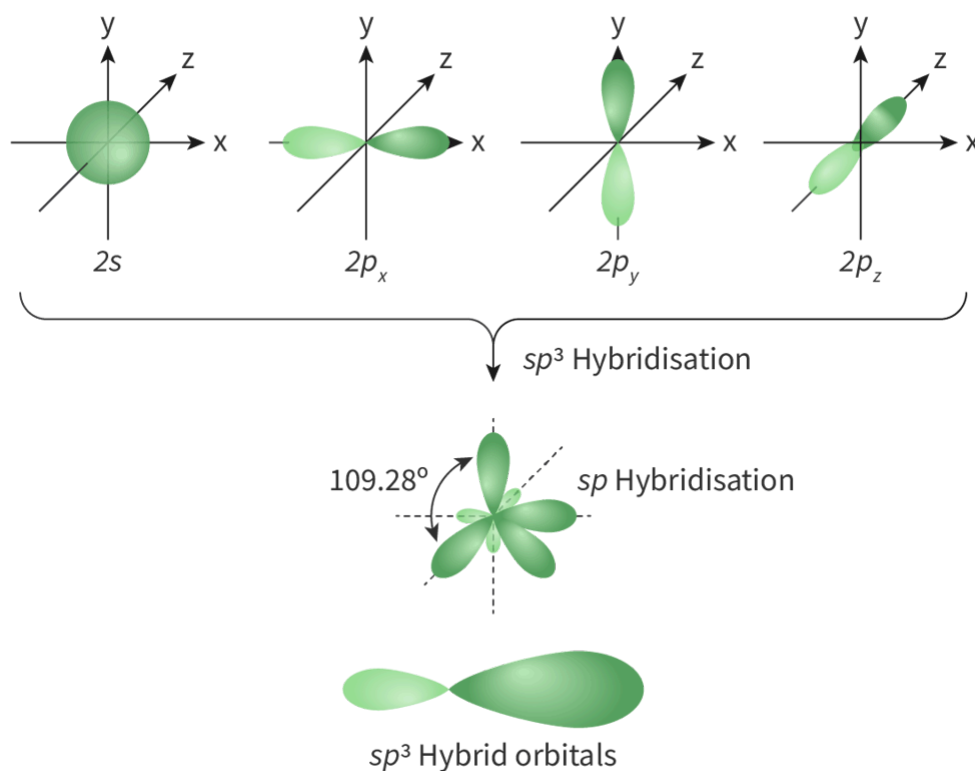


Figure 5. The shape of the sp^3 orbitals reflect the hybridisation.

 More information for figure 5

The diagram illustrates the process of sp^3 hybridisation. It begins with individual 2s and 2p atomic orbitals, which are displayed with their spatial orientations. The 2s orbital is spherical, while the $2p_x$, $2p_y$, and $2p_z$ orbitals have dumbbell shapes oriented along the x, y, and z axes respectively. An arrow indicates the transformation of these orbitals into sp^3 hybrid orbitals, resulting in a new shape with a bond angle of 109.28 degrees, labeled on the

diagram. The final hybrid orbitals have a distinct shape and are shown as a combination of the original orbitals, achieving the tetrahedral geometry typical of sp^3 hybridised compounds.

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The methane molecule, CH_4 , was discussed at the beginning of this section. When carbon forms the four single bonds in methane, it undergoes sp^3 hybridisation, producing four equal orbitals. This hybridised carbon can now form four equal sigma bonds with four hydrogen atoms (**Figure 6**).

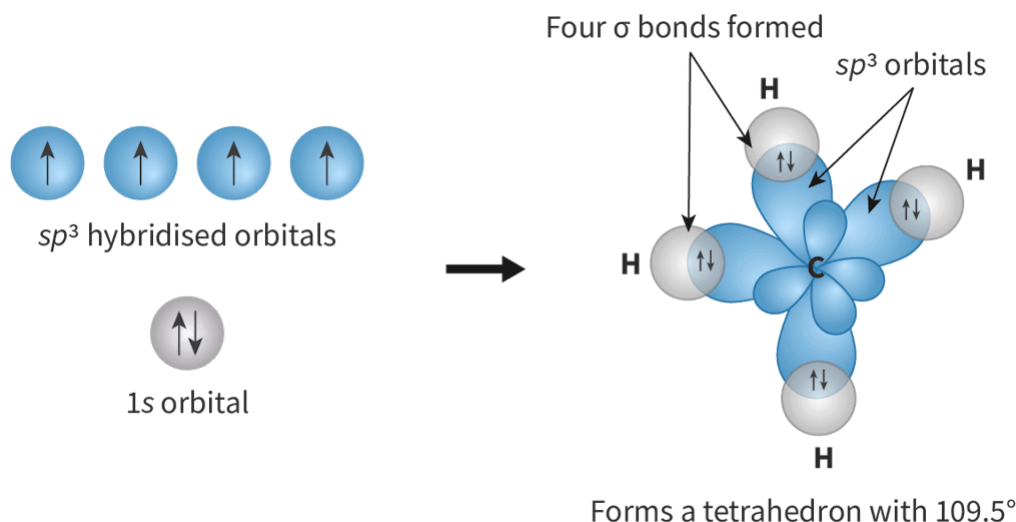


Figure 6. Hybridisation of carbon's orbitals in the methane molecule.

 More information for figure 6

The diagram illustrates sp^3 hybridization in methane. On the left, there are five orbitals: four labeled as " sp^3 hybridised orbitals" with upward arrows and one labeled as "1s orbital" with paired arrows. An arrow points to the right, indicating a transformation. On the right, a methane molecule diagram shows a central carbon atom surrounded by hydrogen atoms forming a tetrahedron. Four " sp^3 orbitals" extend from the central carbon atom, each interacting with a hydrogen atom labeled "H". The text "Four σ bonds formed" is indicated with arrows pointing to each hydrogen bond. The structure forms a tetrahedron with bond angles of 109.5 degrees.

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This hybridisation is called sp^3 as it refers to what is being hybridised: one s-orbital and three p-orbitals (**Figure 7**).

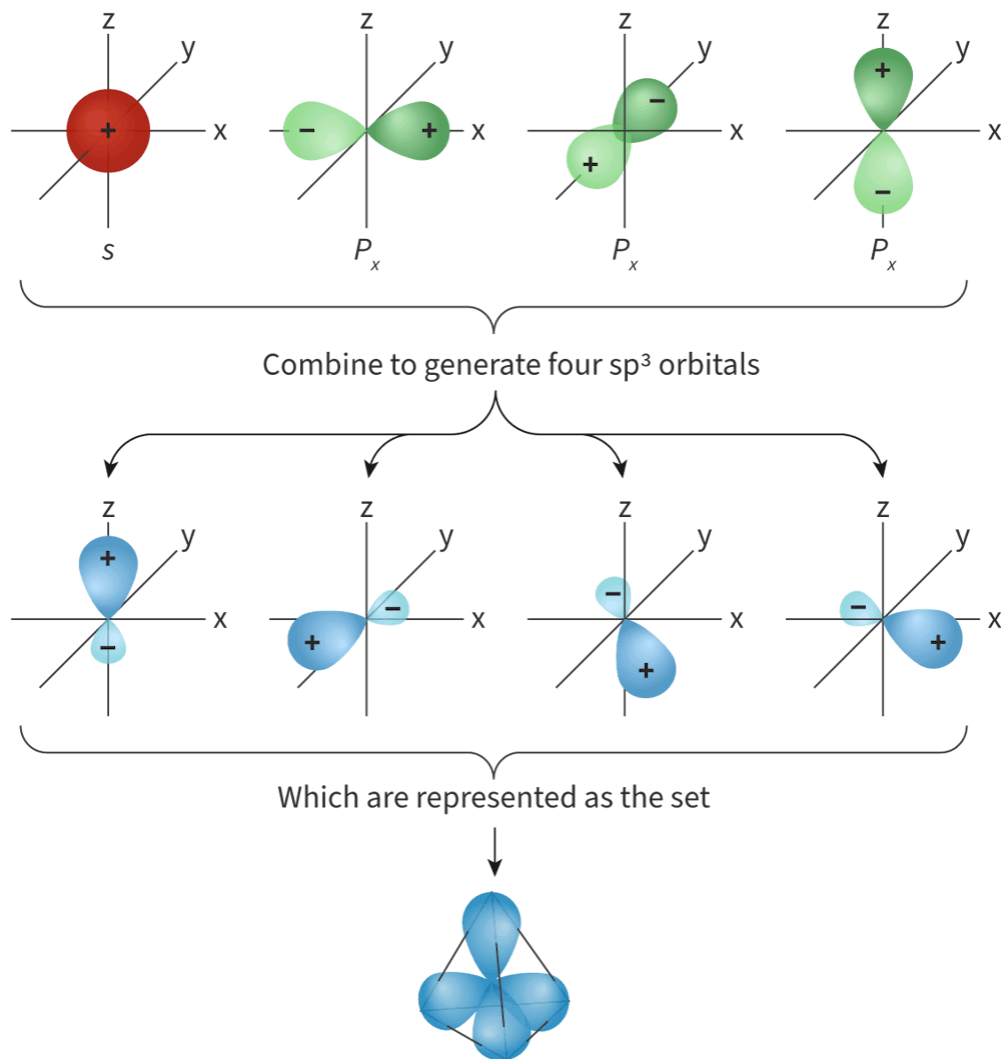


Figure 7. The shape of the orbitals reflects the hybridisation.

 More information for figure 7

The image is a diagram illustrating the process of sp^3 hybridization. At the top, there are representations of separate orbitals: one spherical s orbital and three dumbbell-shaped p orbitals labeled P_x , P_y , and P_z on a set of three axes. The s orbital is colored red, and the p orbitals are colored green. Below these, arrows point downwards, indicating the combination of these orbitals to form four sp^3 hybrid orbitals. These sp^3 orbitals are shown in the middle of the image, each on its own set of xyz axes, with a blue color, depicting a different orientation in space. At the bottom, the four sp^3 orbitals are shown together, forming a symmetrical shape, labeled to indicate that they represent the set of sp^3 hybrids.

[Generated by AI]

sp² hybridisation

Carbon (and other elements) do not always need to form four equal bonds. For example, if a double bond is formed, then it is through a sigma and a pi bond: two different bonds formed by different orbital overlaps. This does not require the same level of hybridisation.

You can see this when carbon forms a double bond, for example in C₂H₄. The orbitals in carbon would undergo sp² hybridisation, one s-orbital and two p-orbitals would become hybridised to produce three equal orbitals (**Figure 8**). One of the 2p-orbitals is left unhybridised.

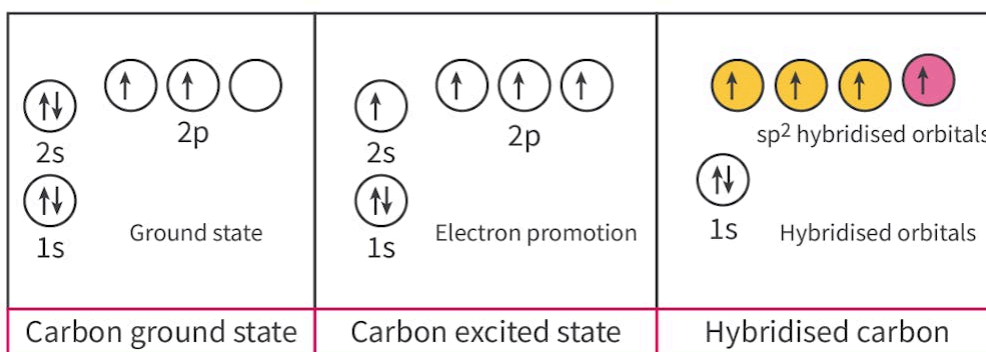


Figure 8. sp² hybridisation of carbon's orbitals.

 More information for figure 8

The image is a labeled diagram illustrating the sp² hybridization process of carbon. It is divided into three sections:

- Carbon ground state:** This section displays the ground state of carbon with filled 1s and 2s orbitals and partially filled 2p orbitals. Two electrons in the 2s are shown with opposite spins, and there are two single electrons in separate 2p orbitals.
- Carbon excited state:** The second section shows electron promotion. One of the electrons from the 2s orbital is promoted to the empty 2p orbital, resulting in one unfilled orbital labeled as 2s and three singly occupied 2p orbitals.
- Hybridised carbon:** The final section shows the formation of sp² hybridized orbitals. One electron in the 1s orbital is unchanged, while the three hybridized orbitals are color-coded, with spins shown in the hybridized orbitals. These represent the new sp² hybridized orbitals formed by mixing one s and two p orbitals, with an unhybridized p orbital maintaining its form.

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The unhybridised 2p-orbital is what will form the pi bond, as part of the double bond. The other three sp^2 hybridised orbitals form sigma bonds.

What do you notice about the shape of the unhybridised p-orbitals in **Figure 9**? Has their shape changed?

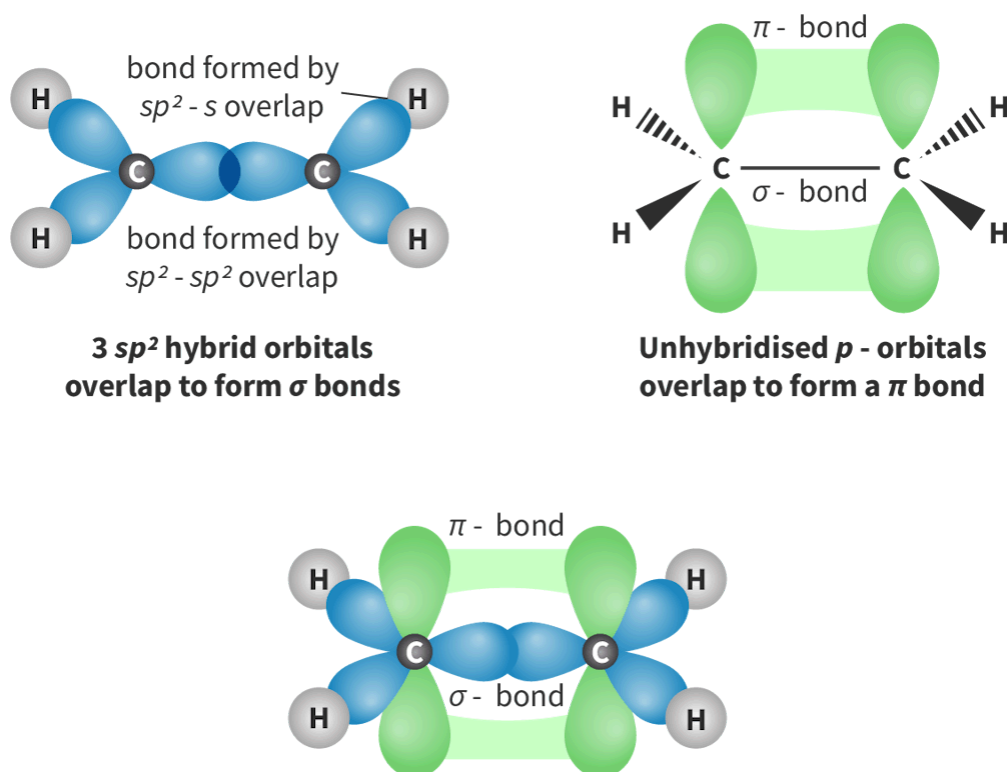


Figure 9. sp^2 hybridisation in ethene, C_2H_4 .

 More information for figure 9

The image is a diagram illustrating the sp^2 hybridisation in ethene, C_2H_4 . It consists of several components:

- Top Left Section:** It shows two carbon atoms bonded together. The carbon atoms, represented by 'C', have blue lobes around them indicating the sp^2 hybrid orbitals. These lobes overlap to form sigma (σ) bonds with hydrogen atoms and each other. It mentions "3 sp^2 hybrid orbitals overlap to form σ bonds."
- Top Right Section:** Depicts a side view of ethene, highlighting the shape of the p-orbitals. The green lobes illustrate the unhybridized p-orbitals forming a pi (π) bond by overlapping sideways above and below the plane of the carbon atoms. It reads "Unhybridised p-orbitals overlap to form a π bond."
- Bottom Section:** A top view showing the same structure with one carbon atom at the center connected to three hydrogen atoms and a carbon atom with blue lobes forming sigma bonds and green lobes forming the pi bond. It labels the σ -bond and π -bond positions.

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The unhybridised p-orbitals that overlap sideways in the pi bond keep the same dumb-bell shape. In the molecule of ethene in **Figure 9**, you can see that only the hybridised orbitals form sigma bonds: the unhybridised p-orbitals are only able to overlap sideways and form pi bonds (**Figure 10**).

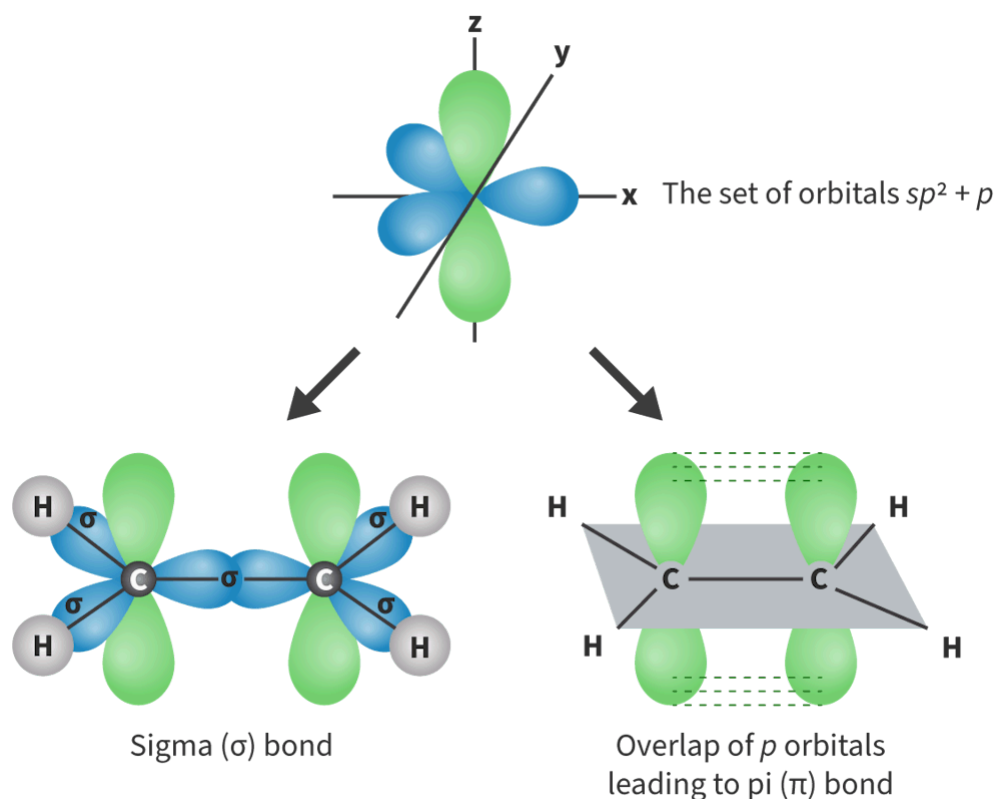


Figure 10. sp^2 hybridisation of carbon's orbitals.

 More information for figure 10

The image is a diagram displaying the sp^2 hybridization of carbon's orbitals. At the top, there is an illustration of a set of $sp^2 + p$ orbitals represented by lobes aligned along the x, y, and z-axes. These lobes are depicted in green and blue, showcasing the geometry of hybridized orbitals.

Below, two structural diagrams are shown: one indicating a sigma (σ) bond formation and the other showing the overlap of p orbitals leading to the formation of a pi (π) bond.

- The first structure is a linear arrangement of a molecule with carbon at the center bonded directly to two hydrogen atoms via sigma bonds. The carbons are connected with sigma bonds, illustrated with overlapping blue lobes.
- The second structure highlights the overlap of p orbitals shown as areas above and below the plane of the sigma bonds, represented with green lobes.

Labels are included to specify each type of bond and orbital involved in the hybridization process, aiding in understanding the distinct components and bonds in the diagram.

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This is because hybridised orbitals all exist on one plane, which makes head-on overlap possible, whereas the unhybridised p-orbital is on a separate plane, only capable of sideways overlap.

sp hybridisation

When carbon forms a triple bond, for example in ethyne, C_2H_2 , it will undergo less hybridisation. It will only hybridise two of its orbitals, one of the s-orbitals and one of the p-orbitals. This produces equal orbitals and is known as sp hybridisation (**Figure 11**). The other two p-orbitals are left unhybridised.

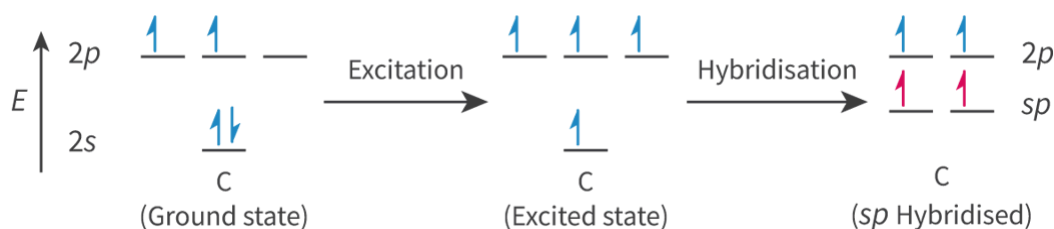


Figure 11. sp hybridisation of carbon's orbitals.

 More information for figure 11

The diagram illustrates the sp hybridisation process of carbon's orbitals in a series of steps. On the left, carbon is in its ground state with two electrons in the $2s$ orbital and two in separate $2p$ orbitals. The diagram shows an upward transition labeled 'Excitation,' leading to an excited state where one electron from the $2s$ orbital is promoted to a $2p$ orbital, making all three $2p$ orbitals singly occupied and the $2s$ depleted.

An arrow labeled 'Hybridisation' points to the right, showing the transformation into the sp hybridised state of carbon. In this state, there's an equal mixing or hybridisation of one s and one p orbital, resulting in two equivalent sp hybrid orbitals and two unchanged $2p$ orbitals. The sp orbitals are depicted as half-occupied with one electron each, while the remaining $2p$ orbitals are shown as singly occupied like before. Each step visually emphasizes the electron movement and orbital transformation during this hybridisation process.

This step-by-step process illustrates how the hybridisation changes carbon's orbital and electronic configuration layout, which is essential in understanding the formation of triple bonds like in acetylene (C_2H_2).

[Generated by AI]

The two hybridised sp-orbitals can form sigma bonds, and the unhybridised p-orbitals can form pi bonds. You can see this in **Figure 12**; the unhybridised p-orbitals on each atom have retained their original dumb-bell shape.

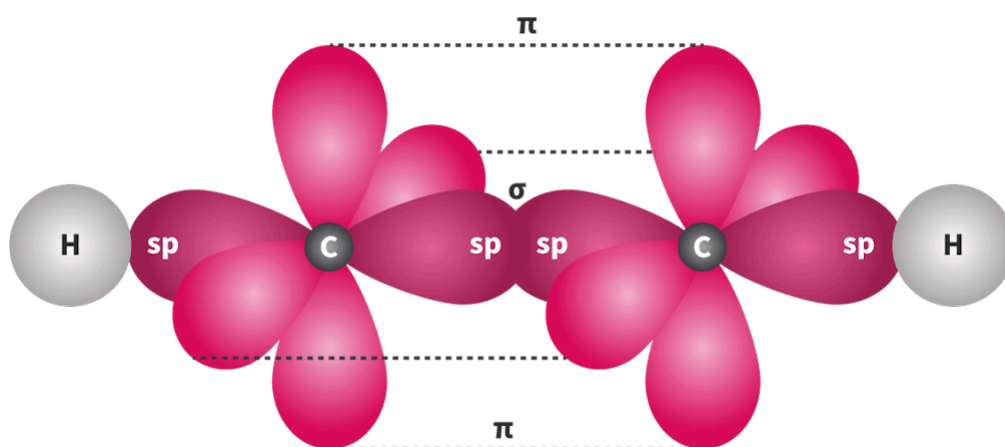


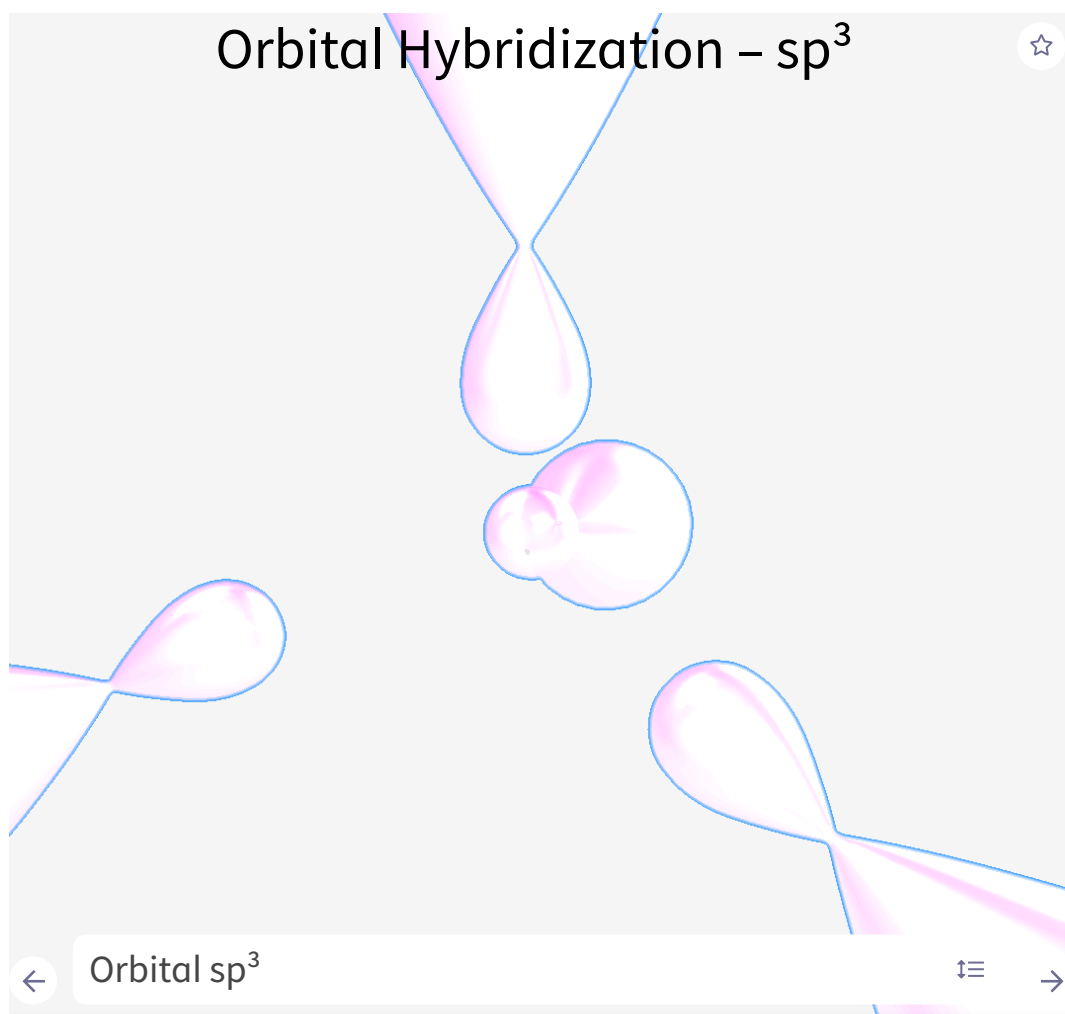
Figure 12. sp hybridisation in ethyne, C₂H₂.

 [More information for figure 12](#)

The diagram illustrates the sp hybridization in an ethyne (C₂H₂) molecule. In the center, two carbon atoms (C) are bonded, each having two sp hybrid orbitals shown in an elongated, oval shape oriented along the axis connecting the two carbon atoms. Between these, a sigma (σ) bond is formed by the overlap of the sp orbitals. From each carbon, perpendicular to the σ bond, are two lobes representing the unhybridized p-orbitals. These lobes create two pi (π) bonds indicated by dashed lines above and below the carbon atoms' axis. At each end of the molecule, a hydrogen atom (H) is bonded to a carbon atom with an sp orbital, signifying a single bond. The p-orbitals maintain a dumb-bell shape and are symmetrical, helping visualize the concept of hybridization.

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Visualising the different types of hybridisation can be difficult for many students. Use these interactive simulations next to explore and imagine this process more easily. Click on the 'introduction' tab at the bottom of each model to learn more about it.



Interactive 1. Corinth Models for sp^3 Hybridisation.

More information for interactive 1

An interactive simulation displays the formation of sp^3 orbital hybridization. Users can rotate and manipulate these representations in any direction. Three 2p orbitals (2px, 2py, and 2pz) and 2s orbital interact and form four sp^3 hybrid orbitals arranged in a tetrahedral structure. Each sp^3 orbital features two lobes, one shorter than the other. The longer lobes are positioned at the corners of the tetrahedron.

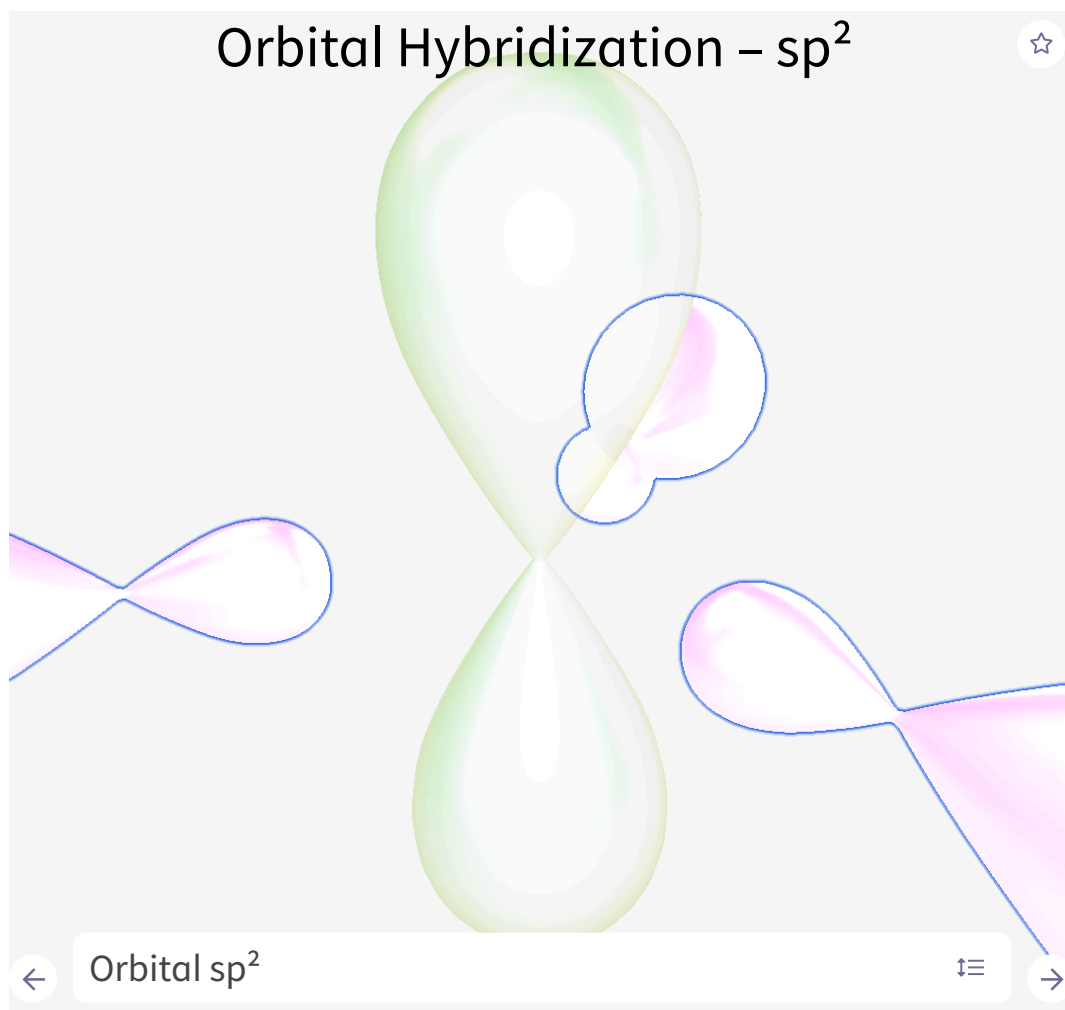
A carousel selector with arrows is at the bottom, allowing switching between the entire model, orbital sp^3 , orbital 2px, orbital 2py, and orbital 2pz. The last four options highlight respective orbitals in the entire model.

Below the selector are options labeled introduction, share, capture, augmented reality, and language. A pause and play button is provided below the model and the option for playing the interactive in loop is given. On the right side, option for changing the background color and option for zooming the interactive are given.

The introduction option provides the following details:

The carbon atom has four valence electrons located in the 2s and 2p orbitals. During sp^3 hybridization, one s-orbital and all three p-orbitals combine to form four equivalent sp^3 hybrid orbitals. These hybrid orbitals are arranged in a tetrahedral geometry around the nucleus, with an angle of $109^\circ 28'$ between them.

This simulation enables users to examine the structure of the sp^3 hybridized orbital.



Introduction Share Capture Augmented Language
Reality

Interactive 1. Corinth Models for sp^2 Hybridisation.

 More information for interactive 1

An interactive simulation displays the formation of sp^2 orbital hybridization. Users can rotate and manipulate these representations in any direction. Two 2p orbitals (2px and 2py) and 2s orbital interact to form three sp^2 hybrid orbitals arranged in a trigonal planar geometry. Each sp^2 orbital features two

lobes, one shorter than the other. The longer lobes are positioned at the corners of the triangular plane. The unhybridized 2pz orbital remains perpendicular to the plane. It has two lobes which are equal in size.

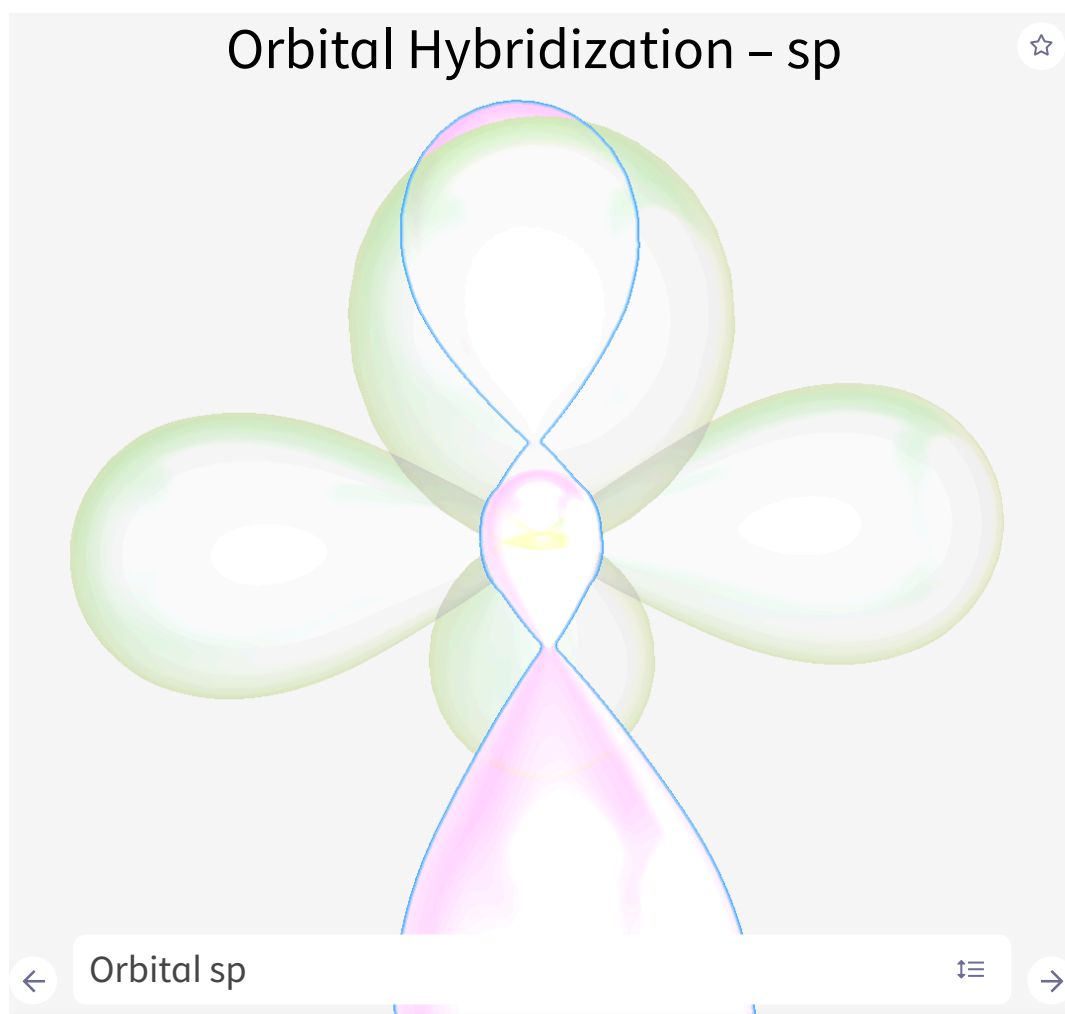
A carousel selector with arrows is at the bottom, allowing switching between the entire model, orbital sp^2 , orbital 2px, orbital 2py, and orbital 2pz. The last four options highlight respective orbitals in the entire model.

Below the selector are options labeled introduction, share, capture, augmented reality, and language. A pause and play button is provided below the model and the option for playing the interactive in loop is given. On the right side, option for changing the background color and option for zooming the interactive are given.

The introduction option provides the following details:

The carbon atom has four valence electrons located in the 2s and 2p orbitals. During sp^2 hybridization, one s-orbital and two p-orbitals combine to form three equivalent sp^2 hybrid orbitals, leaving one p-orbital unhybridized. The sp^2 hybrid orbitals are arranged in a trigonal planar geometry, with their axes pointing toward the corners of an equilateral triangle at 120° angles from each other. The unhybridized p-orbital remains perpendicular to the plane formed by the sp^2 orbitals.

This simulation enables users to examine the structure of the sp^2 hybridized orbital.



Introduction Share Capture Augmented Reality Language

Interactive 1. Corinth Models for sp Hybridisation.

 More information for interactive 1

An interactive simulation displays the formation of sp orbital hybridization. Users can rotate and manipulate these representations in any direction. One 2p orbital (2px) and 2s orbital interact to form two sp hybrid orbitals arranged in a linear geometry. Each sp orbital features two lobes, one shorter than the other. The two unhybridized 2p orbitals (2py and 2pz) remain perpendicular to this linear arrangement. The unhybridized orbital has two lobes that are equal.

A carousel selector with arrows is at the bottom, allowing switching between the entire model, orbital sp, orbital 2px, orbital 2py, and orbital 2pz. The last four options highlight respective orbitals in the entire model.

Below the selector are options labeled introduction, share, capture, augmented reality, and language. A pause and play button is provided below the model and the option for playing the interactive in loop is given. On the

right side, the option for changing the background color and the option for zooming the interactive are given.

The introduction option provides the following details:

The carbon atom has four valence electrons located in the 2s and 2p orbitals. During sp hybridization, one s-orbital and one p-orbital combine to form two equivalent sp hybrid orbitals, leaving the remaining two p-orbitals unhybridized. The sp hybrid orbitals are arranged linearly, 180° apart, along the same axis. The two unhybridized p-orbitals are oriented perpendicular to each other and also perpendicular to the axis of the sp hybrid orbitals. This simulation enables users to examine the structure of the sp hybridized orbital.

Hybridisation of other atoms

To determine the hybridisation of a particular atom in a molecule or ion, you can use the electron domain geometry that is covered in [section S2.2.4](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/>). The electron domain geometry must first be determined using the Lewis formula. **Table 1** shows the relationship between number of electron domains, electron domain geometry and hybridisation of the central atom.

Table 1. The relationship between number of electron domains, electron domain geometry and hybridisation of the central atom.

Regions of electron density	Arrangement	Hybridisation
2	  More information	Linear sp
	The image is a flowchart with a sequence of steps represented by boxes and arrows connecting them. It includes the following steps: "Start the process" - This is the initial step in an oval shape. "Submit documents" - The first action step.	The image symmetric horizontally around the center. The circle labeled horizontally central point. The arc is shown downward at an angle of 180°.

Regions of electron density	Arrangement			
	3. "Review documents" - Decision point where documents are reviewed. 4. "Decision" - A diamond shape indicating a decision is required. It branches into two paths: 5. "Yes" leads to "Approval" in a rectangular box, followed by "Finish process". 6. "No" leads to "Request more info", and then loops back to "Submit documents". 7. "Finish process" - The final step in the process, in an oval shape. The chart illustrates the process of document submission and review, including decision points and outcomes. [Generated by AI]			depicting : arrangement context. [Generated by AI]
3				T r i g o n a l p l a n s p 2

Regions of electron density	Arrangement		Hybridisation	
		Linear	sp	The image showing the central point spaced apart. Each lobe is elongated along the axis extending from the central point. Dotted lines connect the central point to each lobe, showing they are in a straight line. [Generated by AI]
4		Tetrahedral	sp^3	This is a diagram of a tetrahedron, which is a 3D shape with four triangular faces. It represents the arrangement of four electron domains around a central atom. Each lobe represents an electron domain. The tetrahedral arrangement minimizes repulsion between the electron domains. The tetrahedron has four vertices, each representing a region of electron density. The central point is the atom being hybridized. [Generated by AI]

Interconversion between the number of electron domains, the electron domain geometry and the hybridisation is easy:



Figure 13. Interconversion.

 More information for figure 13

The diagram depicts an interconversion process between three concepts: "Number of electron domains," "Electron domain geometry," and "Hybridisation." These components are represented by rectangles. The first rectangle is labeled "Number of electron domains," the middle rectangle is labeled "Electron domain geometry," and the third rectangle is labeled "Hybridisation." Double-headed arrows between the rectangles indicate bidirectional interconversion between these concepts, suggesting a flow of transformation or correlation among them.

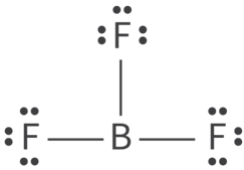
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Determining the hybridisation of an atom:

1. Draw the Lewis formula.
2. Identify the number of electron domains around the central atom.
3. Determine electron domain geometry and molecular geometry using VSEPR theory.
4. Identify the corresponding hybridisation.

Worked example 1

Draw the Lewis formula and state the electron domain geometry, molecular geometry and hybridisation for the boron trifluoride molecule, BF_3 .

Solution steps	Calculations
Step 1: Draw the Lewis formula.	

Solution steps	Calculations
Step 2: Identify the number of electron domains around the central atom.	There are 3 electron domains around boron.
Step 3: Determine electron domain geometry using VSEPR theory.	Three electron domains correspond to trigonal planar geometry.
Step 4: Determine the molecular geometry using VSEPR theory.	Three bonding electron domains and no non-bonding electron domains correspond to trigonal planar molecular geometry.
Step 5: Identify the corresponding hybridisation.	Since there are 3 electron domains, the B atom will form 3 hybrid orbitals. The B atom will therefore undergo sp^2 hybridisation.

Worked example 2

Draw the Lewis formula and state the electron domain geometry, molecular geometry and hybridisation for the ammonia molecule, NH_3 .

Solution steps	Calculations
Step 1: Draw the Lewis formula.	$\begin{array}{c} \text{H} - \ddot{\text{N}} - \text{H} \\ \\ \text{H} \end{array}$
Step 2: Identify the number of electron domains around the central atom.	There are 4 electron domains around nitrogen.
Step 3: Determine electron domain geometry using VSEPR theory.	Four electron domains correspond to tetrahedral

Solution steps	Calculations
Step 4: Determine the molecular geometry using VSEPR theory.	Three bonding electron domains and one non-bonding electron domain result in trigonal pyramidal geometry.
Step 5: Identify the corresponding hybridisation.	Since there are 4 electron domains, the N atom forms 4 hybrid orbitals. The N atom will therefore be sp^3 hybridised.

Worked example 3

Draw the Lewis formula and state the electron domain geometry, molecular geometry and hybridisation for the carbon dioxide molecule, CO_2 .

Solution steps	Calculations
Step 1: Draw the Lewis formula.	$\ddot{O} = C = \ddot{O}$

Solution steps	Calculations
Step 2: Identify the number of electron domains around the central atom.	There are 2 electron domains around carbon.
Step 3: Determine electron domain geometry using VSEPR theory.	Two electron domains correspond to linear geometry.
Step 4: Determine the molecular geometry using VSEPR theory.	Two bonding electron domains and no non-bonding electron domains correspond to linear geometry.
Step 5: Identify the corresponding hybridisation.	Since there are 2 electron domains, the C atom uses sp hybrid orbitals to form 2 hybrid orbitals. The C atom p -orbitals: $s + p = sp$ hybridised.

Study skills

Remember that sigma bonds and non-bonding electrons are held in hybrid orbitals, but pi bonds are not and that electron domain geometry is used to determine the hybridisation.

Hybridisation in molecules with more than one central atom

In VSEPR theory, the fact a molecule has more than one central atom may mean it has more than one molecular geometry present in it (see [section 2.2.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/>)). Remember that when we determine the molecular geometry, it is just for that particular central atom with that specific number of electron domains.

Similarly, when there is a molecule with more than one central atom there may also be more than one particular type of hybridisation present in the molecule. For example, in a molecule of propene, CH_2CHCH_3 , (**Figure 14**) there are three central carbon atoms, there are two different electron domain geometries and therefore two different hybridisations.

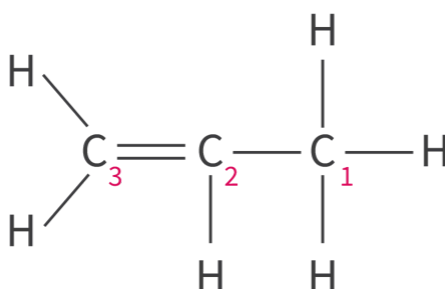


Figure 14. Propene.

 More information for figure 14

The image is a structural diagram of the propene molecule. It consists of three central carbon atoms, each labeled with numbers: 1, 2, and 3. The first carbon atom (labeled 1) is bonded to three hydrogen atoms and is oriented in a tetrahedral configuration, indicating sp^3 hybridization. Carbons 2 and 3 are connected by a double bond, giving them a trigonal planar shape and sp^2 hybridization. Each of these carbon atoms is also associated with one hydrogen atom. The configuration of bonds and orientation indicates the electron domain geometry and hybridization of each central atom in the propene molecule.

[Generated by AI]

The carbon atom labelled 1 has a tetrahedral electron domain geometry as it has four electron domains. As a result, it has sp^3 hybridisation. The carbon atoms labelled 2 and 3 are trigonal planar as they each have three electron domains. The hybridisation of both carbon atoms is therefore sp^2 .

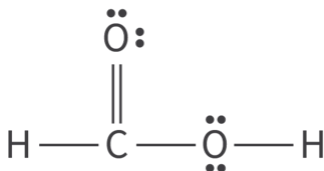
Study skills

When you are determining the hybridisation present in a molecule with more than one central atom you must treat each central atom separately to determine its electron domain geometry using VSEPR theory and its hybridisation.

Worked example 4

Determine the hybridisation of each central atom in molecules of:

1. Methanoic acid, HCOOH

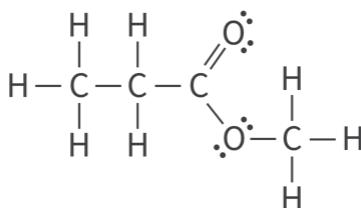


 More information

The image is a molecular diagram of methanoic acid (HCOOH). It shows a carbon (C) atom double-bonded to an oxygen (O) atom at the top, which has two lone pairs of electrons. The carbon atom is also single-bonded to another oxygen atom on the right, which also has two lone pairs of electrons, and this oxygen is additionally bonded to a hydrogen (H) atom. At the bottom left, the carbon is single-bonded to another hydrogen atom.

[Generated by AI]

2. Methyl propanoate, $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$.



 More information

1. Methanoic acid, HCOOH

Solution steps	Calculations
Step 1: Draw the Lewis formula.	This was already drawn H —
Step 2: Identify the number of electron domains around the central atoms.	There are 4 electron dc H —

Solution steps	Calculations
	There are 3 electron do H —
Step 3: Determine electron domain geometry using VSEPR theory.	Four electron domai Three electron domai
Step 4: Identify the corresponding hybridisation.	Four electron domai hybridised Three electron domai

2. Methyl propanoate, $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$.

Solution steps	Calculations
Step 1: Draw the Lewis formula.	This was already drawn $\begin{array}{c} \text{H} \\ \\ \text{H} - \text{C} \\ \\ \text{H} \end{array}$

$$\begin{array}{c} \text{H} \\ | \\ \text{H} - \text{C} \\ | \\ \text{H} \end{array}$$

Solution steps	Calculations
Step 3: Determine electron domain geometry using VSEPR theory.	Four electron domain Three electron domain
Step 4: Identify the corresponding hybridisation.	Four electron domain hybridised Three electron domain

Hybridisation in resonance hybrid molecules

In section 2.2.12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/benzene-hl-id-45135/>) we discussed the resonance hybrid structure of benzene, seen in **Figure 15**.

The molecule has a six-carbon hexagonal ring structure, with each carbon single bonded to its adjacent carbon, and three additional pairs of delocalised electrons spread across the entire ring.

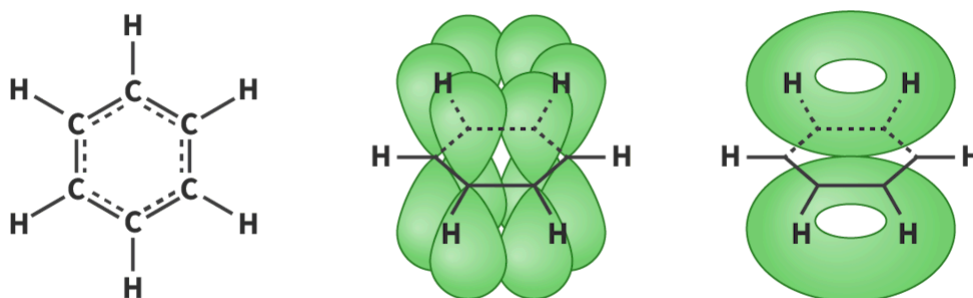


Figure 15. The delocalised electrons in a benzene molecule.

 More information for figure 15

The image shows three representations of a benzene molecule from left to right. The first diagram displays a traditional hexagonal ring of six carbon atoms, each bonded to a hydrogen atom with alternating single and double

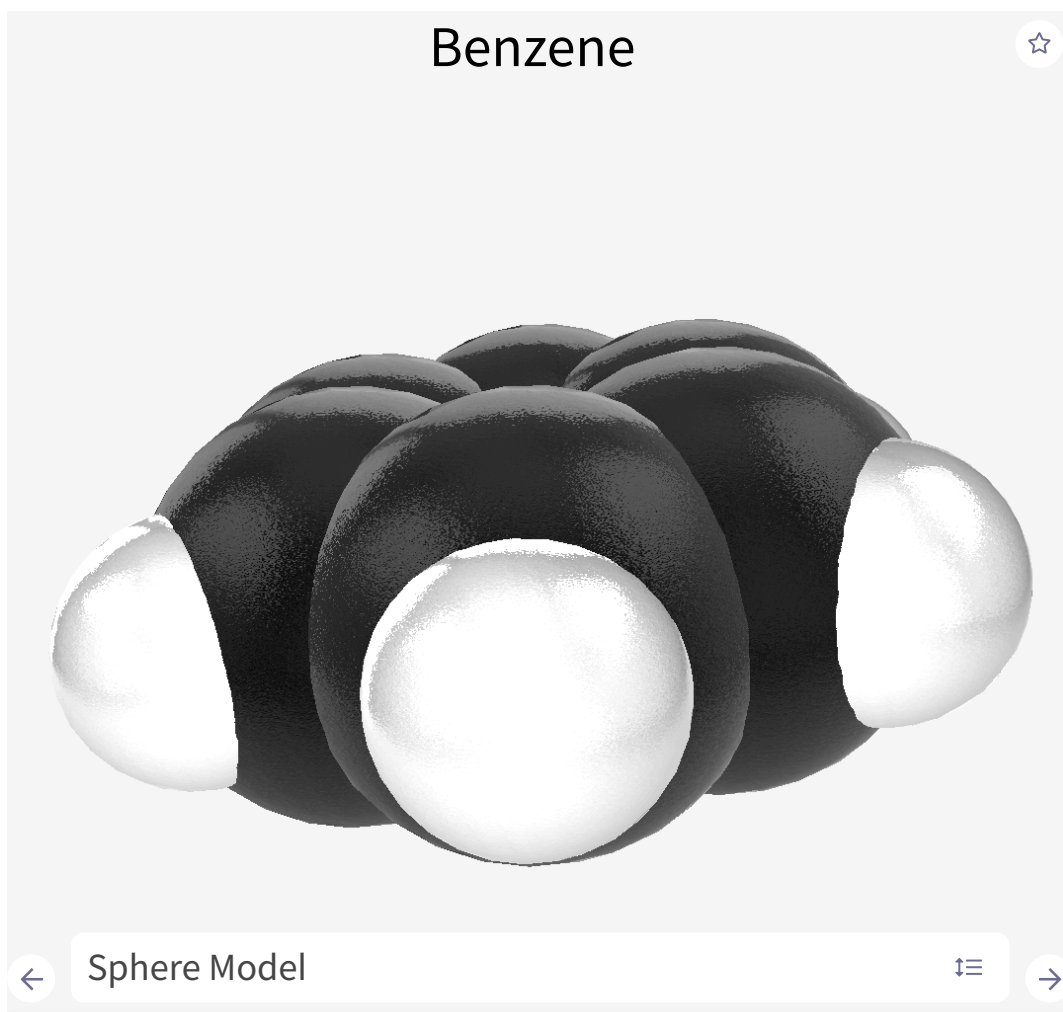
bonds represented by dashed lines inside the ring. This illustrates the conventional Lewis structure of benzene.

The middle diagram presents a more complex view demonstrating the concept of delocalized electrons. In this representation, the carbon atoms are bonded to the hydriens as before, but the focus is on the overlapping p-orbitals throughout the ring. This illustrates the electron cloud of delocalized π -bonds that extend above and below the plane of the carbon atoms, forming a continuous loop.

The third diagram shows a side view of the benzene molecule pi electron cloud. It highlights the torus shape of electron clouds above and below the plane of the carbon ring, emphasizing how these π -electrons are delocalized over all six carbon atoms in the benzene molecule. Each carbon is bonded to hydrogen atoms, similar to previous representations, but the focus remains on the three-dimensional shape of the electron clouds.

[Generated by AI]

The simulation in [section S2.2.12 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/benzene-hl-id-45135/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/benzene-hl-id-45135/) gave you a visualisation the structure of the benzene model, C_6H_6 . Selecting the 'Electron system π ' showed the resonance hybrid structure. Revisit this simulation again now and use it to understand which electrons become delocalised.

[Introduction](#)[Share](#)[Capture](#)[Augmented
Reality](#)[Language](#)

Interactive 2. 3D Structure of the Benzene Molecule.

 More information for interactive 2

An interactive, three-dimensional illustration displays different representations of benzene. Users can rotate these representations in any direction. A carousel selector with arrows is at the bottom, allowing switching between a ball-and-stick model, a sphere model, and a structural formula.

In the ball-and-stick model, six large spheres representing carbon atoms are bonded in a hexagonal structure with alternating single and double bonds. Each carbon atom is connected to a smaller sphere representing hydrogen by a single bond. The bonds are represented as thin rods. In this model, the carbon and hydrogen atoms, pi electron system, and molecular bonds can be visualized by choosing the option.

In the sphere model, six large spheres representing carbon atoms are fused in a hexagonal pattern. Each carbon atom is fused with a smaller sphere representing a hydrogen atom on the outside. In this model, the carbon and hydrogen atoms can be visualized by choosing the options.

The structural formula features a hexagonal arrangement with a CH occupying each corner and alternating single and double bonds between them.

On the right side, the option for changing the background color and the option for zooming the interactive are given.

Below the selector are options labeled introduction, share, capture, augmented reality, and language.

The introduction option provides the following details:

Molecular formula: C_6H_6

Molar mass: 78.11 g/mol

Description: Benzene is an aromatic, colorless liquid with a sweet odor. It is highly toxic and carcinogenic. When burned, it produces large amounts of soot. Benzene vapors form an explosive mixture with air upon ignition.

Applications: Benzene is used as a solvent for many organic substances and is an important raw material in the production of dyes and pharmaceuticals.

These three-dimensional representations allow users to explore benzene structure, including its bonding and geometry. They provide a clear view of the spatial arrangement of atoms in the molecule.

Can you see why benzene is best described as having a ring of delocalised π electrons?

As both single, double and triple bonds are all counted as one electron domain, the presence of delocalised electrons does not change how many electron domains surround each carbon, each carbon still has three electron domains. Therefore, the hybridisation experienced by carbon in benzene is sp^2 (**Figure 16**).

This can also be determined using the sigma and pi bonds. Each carbon-hydrogen bond is a sigma bond and each carbon-carbon bond contains a sigma bond. The double bonds also contain a pi bond. We have established previously that pi bonds cannot form from hybridised orbitals, so one of the p-orbitals on each carbon must remain unhybridised. Three hybridised orbitals correspond to sp^2 .

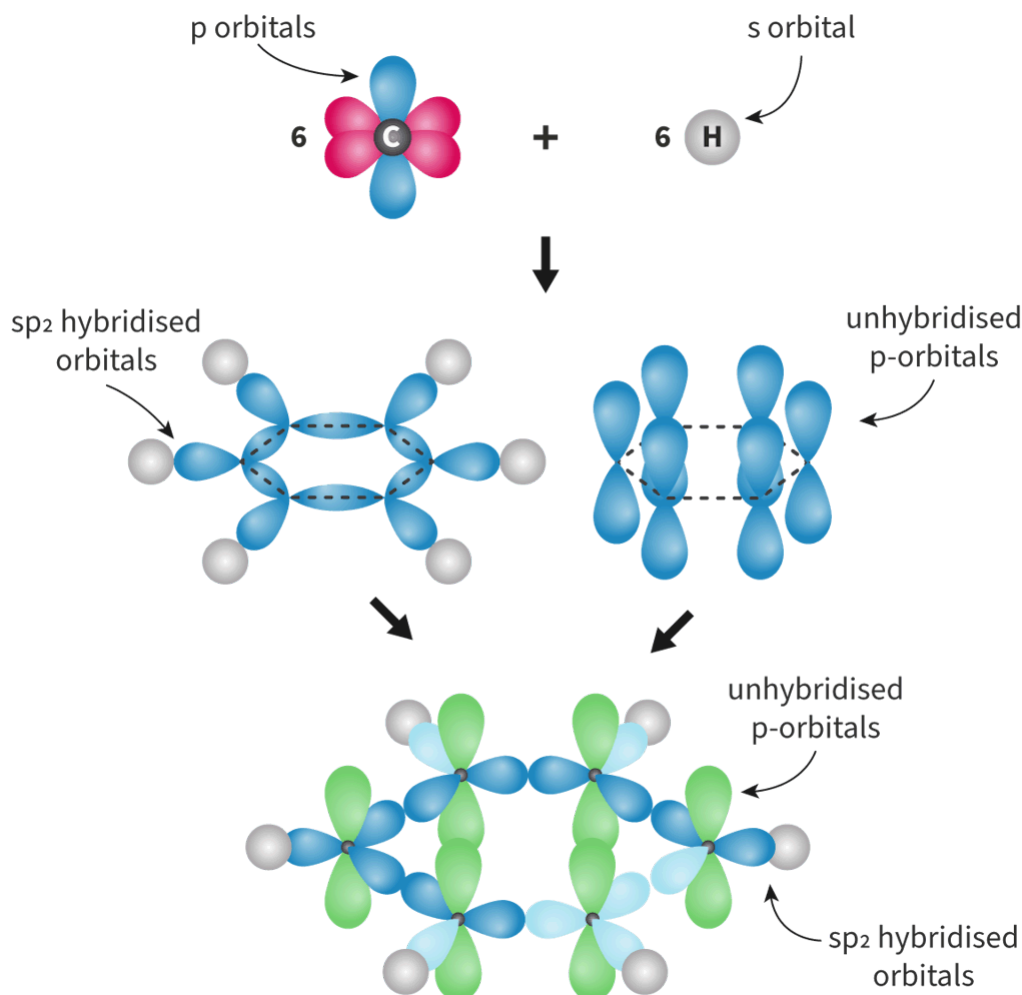


Figure 16. Hybridisation in benzene.

 More information for figure 16

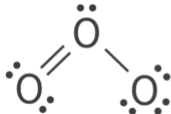
The diagram illustrates the hybridisation process in benzene. It shows the transformation of atomic orbitals into molecular orbitals during the formation of benzene. At the top, six carbon p orbitals and six hydrogen s orbitals are shown. This is followed by a combination to form sp^2 hybridised orbitals and unhybridised p-orbitals.

- At the top left, there are six p orbitals labeled "p orbitals," with a carbon atom (labeled C) at the center and pink lobes.
- On the top right, six hydrogen s orbitals are labeled "s orbital."
- Arrows indicate the interaction between these orbitals.
- Below, two sets of orbitals are displayed:
 - On the left, sp^2 hybridised orbitals, comprising three blue orbitals connected by small white spheres.
 - On the right, unhybridised p-orbitals with three blue pairs aligned vertically, with dashed lines indicating pi bonds.
- Finally, at the bottom, a complex arrangement shows a combination of unhybridised p-orbitals and sp^2 hybridised orbitals, depicting benzene's structure with alternating green and blue lobes and small white spheres labeled " sp^2 hybridised orbitals" and "unhybridised p-orbitals."

[Generated by AI]

Worked example 5

Determine the hybridisation of the central oxygen atom in a molecule of ozone, O_3 .

Solution steps	Calculations
Step 1: Draw the potential Lewis formulas for this resonance hybrid molecule.	
Step 2: Identify the number of electron domains around the central atom.	There are 3 electron domains around the central oxygen.
Step 3: Determine electron domain geometry using VSEPR theory.	Three electron domains
Step 4: Identify the corresponding hybridisation.	Three electron domains hybridised

This activity will allow you to apply the skills introduced in the last section to some unfamiliar molecules.

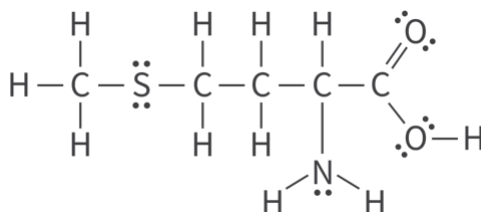
Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Questions

1. Methionine, $\text{CH}_3\text{SCH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{CO}_2\text{H}$, is an amino acid found in proteins.

The Lewis formula of this compound is shown below.



More information

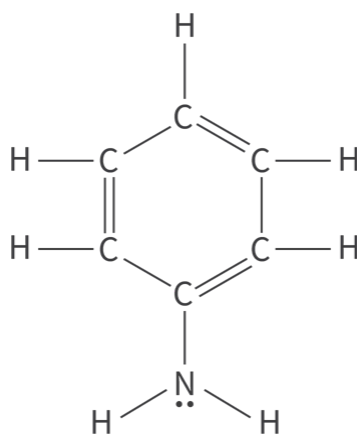
The image is a structural diagram of a chemical compound molecule. The structure includes a sequence of elements linked by single, double, and lone pairs of electrons. Starting from the left, a hydrogen atom (H) is bonded to a carbon atom (C) which is bonded to a sulfur atom (S) with two lone pairs. This sulfur atom is then linked to another carbon atom, followed by a chain of more carbon atoms, each bonded with hydrogen (H) atoms. On the right side of the structure, there is a carbon atom double-bonded to an oxygen atom at the top and single-bonded to an oxygen atom on the right, which in turn is bonded to a hydrogen atom, forming a carboxylic acid group. Additionally, a nitrogen atom, bonded with a single hydrogen atom, is attached to the middle carbon, showcasing an amine group. The diagram represents a basic organic chemistry molecule, indicating the specific atom connectivity and chemical structure.

[Generated by AI]

1. (a) Identify the total number of sigma and pi bonds present in the molecule.
(b) Describe the molecular geometry of all carbon (there may be more than one!), sulfur, nitrogen and oxygen atoms.
(c) Determine the hybridisation type of each carbon, oxygen, and the

nitrogen and the sulfur.

2. Phenylamine, commonly known as aniline, is an aromatic amine . The Lewis formula of aniline is shown here.



 More information

This image is a chemical structure diagram of pyridine. It depicts a hexagonal benzene ring consisting of six carbon atoms. The carbon atoms are connected by alternating single and double bonds. Attached to the top right carbon atom is a single hydrogen atom, while the lower right and upper left carbon atoms each have hydrogen atoms bonded to them as well. The nitrogen atom (N) replaces one of the carbon atoms in the benzene ring at the bottom vertex of the hexagon and has a lone pair of electrons shown as two dots. The nitrogen atom is bonded to a hydrogen atom externally, forming an NH group.

[Generated by AI]

1. (a) Identify the total number of sigma and pi bonds present in the molecule.
(b) Deduce the electron domain and molecular geometries present for the carbon and nitrogen atoms in the molecule.
(c) Determine the hybridisation of the carbon and nitrogen atoms in aniline.
(d) Predict and justify the bond length and strength of the carbon—carbon bonds in aniline.
(e) Explain what is meant by the terms ‘delocalisation’ and ‘resonance’.

1. (a) There are 19 sigma bonds and 1 pi bond in the molecule.
 (b) All C atoms in the molecule have a molecular geometry of tetrahedral, except for the C atom that is double bonded to the O atom, which has a molecular geometry of trigonal planar. The S atom has a molecular geometry of bent/angular; the N atom is trigonal pyramidal. The O atom with a single bond to the C atom is bent/angular. The O atom with a double bond to the C atom would be undefined, since it is only bonded to one other atom (three atoms are required to make a bond).
 (c) All C atoms in the molecule have a hybridisation of sp^3 , except for the C atom that is double bonded to the O atom, which has an sp^2 hybridisation. The S and N atoms are sp^3 hybridised. The O atom with a single bond to the C atom is sp^3 hybridised. The O atom with a double bond to the C atom is sp^2 hybridised.
2. (a) There are 14 sigma bonds and 3 pi bonds in the molecule.
 (b) All C atoms in the molecule have an electron domain and molecular geometry of trigonal planar. The N atom has a tetrahedral electron domain geometry and a trigonal pyramidal molecular geometry.
 (c) All C atoms in the molecule have a hybridisation of sp^2 . The N atom has an sp^3 hybridisation.
 (d) Using [section 1.3.11](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-or-average-covalent-bond-lengths-id-45113/>) of the DP chemistry data booklet, the C—C bond length would be approximately 140×10^{-12} m. The length of a C—C bond is 154×10^{-12} m and a C=C bond is 134×10^{-12} m. Since these bonds have a bond order of 1.5, the length would be between that of a C—C and C=C bond. Similarly, using [section 1.3.12](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) of the DP chemistry data booklet, the strength of the C—C bonds would be approximately 480 kJ mol^{-1} . A C—C has a bond enthalpy of 346 kJ mol^{-1} and a C=C bond has a bond enthalpy of 614 kJ mol^{-1} .
 (e) Delocalisation refers to an electron that is not associated with one specific atom and is free to move within the molecular structure. Resonance refers to a molecule in which more than one valid Lewis formula can be drawn, due to the presence of delocalised electrons.